



Lab Report: Single Slit Diffraction

Student Name: _____

Date: _____

1. Objective

To observe the diffraction pattern produced by a single slit and determine how slit width (a) influences the spread of light.

2. Theory

Minima in a single slit pattern occur at angles θ where:

$$a \sin \theta = m\lambda \quad (m = 1, 2, 3\ldots)$$

3. Data Collection (Set Wavelength $\lambda = 650 \text{ nm}$)

Trial	Slit Width (a)	Central Max Width (W)	Observed Pattern Description
1	0.04 mm		
2	0.08 mm		
3	0.16 mm		

4. Analysis Questions

- Based on your data, what happens to the width of the central maximum as the slit becomes narrower?
- Calculate the angle θ for the first minimum ($m = 1$) for Trial 1.